

S4I Expression of Interest (EOI) for Flagship Scaling Solution Tracks

Section 0 – Title and contact Information

- **Title of the proposed solution track**
Scaling climate-smart water and soil solution bundles through participatory and inclusive innovations in Guatemala and Colombia.
- **Title of the AoW2 Flagship that the solution track aligns with**
1.2 Scaling Smart Irrigation and Climate-smart Resilient Water Systems.
- **Lead Scientist(s) / Contact Person(s) / Center(s)**
Deissy Martínez-Baron, Alliance of Bioversity and CIAT
- **Emails of the Lead Scientist(s) / Contact Person(s) / Center(s)**
d.m.baron@cgiar.org
- **Key CGIAR Center Partners (in the case of multi-Center Eols)**
Potentially work with IWMI.
- **Technical link to other Science Programs – name the corresponding Science Programs by name**
Climate Action SP, potentially DTA.

Acronyms

Acronym	Name
ADIPAZ	Asociación de Desarrollo Integral de la Parte Alta de Zacapa
ASORECH	Asociación Regional Campesina Ch'ortí
CDRO	Asociación de Cooperación para el Desarrollo Rural de Occidente
COOSAJO	Cooperativa de Ahorro y Crédito Integral “San José Obrero”
CSV / CSVs	Climate-Smart Villages
FEDECOVERA	Federación de Cooperativas de Alta Verapaz
GCF	Green Climate Fund
IADB	Inter-American Development Bank
ICTA	Instituto de Ciencia y Tecnología Agrícola de Guatemala
IFI	International Financial Institution
IWMI	International Water Management Institute
KPIs	Key Performance Indicators
LTACs	Local Technical Agroclimatic Committees
MADR	Ministerio de Agricultura y Desarrollo Rural
MADS	Ministerio de Ambiente y Desarrollo Sostenible
MAGA	Ministerio de Agricultura, Ganadería y Alimentación
MARN	Ministerio de Ambiente y Recursos Naturales
MCEC	Movimiento Cívico de Extensionistas de Colombia
NDCs	Nationally Determined Contributions
PICSA	Participatory Integrated Climate Services for Agriculture
UNDRR	United Nations Office for Disaster Risk Reduction
UNEP/GRID-Geneva	United Nations Environment Program / Global Resource Information Database – Geneva

Section 1 – Innovation Package, Value Proposition & Alignment

Purpose: To establish what the innovation is, who the customers and end-users are, why it matters, and how it connects to S4I's Flagships and Grand Challenges.

- **The problem to be solved:** Describe the challenge you are seeking to overcome through the proposed Scaling Solution track. What country / countries will the Solution Track operate in? Be sure to clarify who you are seeking to benefit with this work, with attention paid to gender and social inclusion. (~150 words)

Farmers in Guatemala's Dry Corridor and in Colombia face increasingly severe climate risks, including prolonged droughts, erratic rainfall, soil degradation, and declining water availability for food production. In Guatemala, droughts reduced staple crop yields by up to 25% in 2023 (UNDRR, 2024), while soil erosion reaches 66 million tons annually (UNEP/GRID-Geneva, n.d.). In Colombia, floods during the 2010–2011 La Niña episode affected 3.5 million hectares and over three million people, and climate models project rising temperatures of 3–5°C and rainfall reductions of up to 30% by mid-century, threatening key crops such as maize, rice, and potatoes (IPCC, 2022; GCF, 2022). With over 90% of agriculture rain-fed (MADR, 2021), these risks undermine productivity, food security, and rural livelihoods.

Although climate-smart soil and water practices exist, their adoption remains limited and fragmented. This Solution Track will operate in Guatemala and Colombia to overcome this gap by reaching smallholder farmers, cooperatives, and rural communities, especially women, youth, and marginalized groups, through inclusive innovation networks that enable access to context-specific solutions, reduce vulnerability, and build adaptive capacity at scale.

- **Your scaling ambition:** Describe the ambition of your proposed Scaling Solution track. Which of the KPIs do you believe you will contribute most towards, and at what level of quantitative and qualitative impact? (~150 words)
 - **Short-term (by 2026):** Over **20,000 farmers** engaged with climate-smart water–soil solution bundles, with **40% women and youth participation**. The innovation network grows to more than **12** nationally and regionally **partners, reporting strengthened knowledge and capacities** on water–soil conservation, efficiency, and climate risk management.
 - **Medium-term (2027–2028):** Approximately **80,000 farmers (50% women and youth)** in Guatemala and Colombia reached through inclusive training, digital platforms, and information services, leveraging PICSA, LTACs, and CSVs. Over **15 organizations** collaborate in innovation networks, advancing bundles at both plot and watershed scales. Expected outcomes include at least a **20% reduction in crop failure** and **one policy informed by or integrating innovation bundles**.

This track contributes most directly to **KPIs on the adoption of CGIAR innovations and policy uptake**, while also strengthening institutional capacities.

- **Innovation & Value Proposition** – Describe the innovation or bundle or package (maturity, novelty, suitability for scaling) and the benefits it delivers to institutional partners/customers and end-users (e.g., productivity, resilience, nutrition, risk reduction, institutional capacity). Can include linkages to other science programs that will collaborate and work on the innovation/bundle/package (~150 words)

The proposed innovation introduces climate-smart water and soil solution bundles through participatory approaches in Guatemala and Colombia. These bundles integrate context-specific practices and tools to improve soil health, optimize water use, and strengthen smallholder resilience to climate variability. Their novelty lies not only in combining validated practices but also in their delivery with champion farmers, community organizations, and financial institutions, ensuring scalability across diverse agroecological and social contexts. With a Scaling Readiness level of 9 and Use level of 8, the innovation demonstrates maturity and proven impact under field conditions. For institutional partners, the bundles provide an adaptable, evidence-based package that advances their mission and builds credibility with farmers by promoting scientifically proven innovations that can be used in program planning and investments. For end-users, the bundles increase productivity, reduce climate risks, and improve food and nutrition security while fostering sustainable resource management and inclusive participation. Collaboration with the Climate Action SP and the Digital Transformation Accelerator will further strengthen participatory innovations and digital tools for scaling.

- **Customers & Alignment** – Identify the main institutional customers (ministries, donor programs, agribusinesses, lenders) and end-users (farmers, SMEs, cooperatives). Show how the track aligns to an AoW2 Flagship and Grand Challenge. (~100 words)

In Guatemala, institutional customers include MAGA, MARN, and ICTA, supported by local partners such as CDRO, ADIPAZ, and ASORECH. Financial cooperatives such as COOSAJO and FEDECOVERA expand access to credit, while Global Communities and Agropecuaria Popoyán strengthen outreach, and LTACs deliver climate services to farmers and cooperatives. In Colombia, MADR and MADS provide enabling frameworks, while networks such as the MCEC, Asopro Orgánicos, and 28 LTACs connect innovations with smallholders and cooperatives. Lenders such as Contactar expand financial access, ensuring institutional customers and end-users—farmers, SMEs, and cooperatives—benefit from inclusive innovations aligned with national priorities.

This scaling track directly contributes to Flagship 1.2 Scaling Smart Irrigation and Climate-smart Resilient Water Systems, aligning with Grand Challenge 1 by closing the adaptation gap through scalable smart water management and climate services, combining soil and water innovations with climate information, strengthening institutional capacity, and enabling inclusive, context-specific solutions that build resilience and reduce risks.

- **Evidence of Impact** – Provide citations and evidence of both effectiveness (impact) and scalability in practice (uptake beyond pilots, adoption drivers). (~150 words)

Evidence from Guatemala and Colombia shows that climate-smart soil and water solution bundles deliver measurable impacts on productivity, resilience, and reduced climate vulnerability. In Guatemala, over 60% of farmers in the CSV adopted at least one practice, reporting yield gains up to 40% in maize and beans, improved water availability in dry seasons, and 70% perceiving reduced vulnerability, mainly through lower drought losses and faster recovery (Martínez, J.D. et al., 2024). Cost-benefit analyses show positive ratios up to 3.4 for terraces, live barriers, and community reservoirs (Gustavo Sain et al., 2016). In Colombia, 56% of CSV households implemented CSA practices; 80% adopted drought-resistant beans, 63% organic fertilizers, and 73% climate-adapted home gardens. Between 70–85% of farmers reported higher income, food security, and reduced climate vulnerability, linked to technologies adapted to water variability (Almenteros H. et al., 2019). CSA Country Profiles confirm reductions in erosion (up to 35%), improved soil fertility, moisture, and water-use efficiency, plus co-benefits such as lower emissions in silvopastoral systems, diversification income, and integration into national strategies (World Bank, CIAT & CATIE, 2014).

Section 2 – Scaling Pathway, Sustainability, Partnerships & Risks

Purpose: To assess whether there is a credible pathway for the innovation to scale through public, private, or PPP delivery models — supported by sustainability plans, partner demand and commitments, financial leverage, and risk mitigation.

- **Scaling Pathway & Sustainability** – Describe how the innovation bundle and/or package will scale (public, private, PPP). Explain how the solution will remain viable at scale (revenues, public financing, PPP, IFI/donor support), with milestones or cost-recovery targets. (~250 words)

The scaling of climate-smart water and soil solution bundles in Guatemala and Colombia will follow a public-private-community partnership (PPP) pathway. National ministries (MAGA, MARN, MADR, MADS) already have enabling public policy frameworks, extension systems, and financing mechanisms, while local cooperatives, SMEs, and agribusinesses (e.g., COOSAJO, FEDECOVERA, Popoyán, MCEC, Contactar) will ensure service delivery, credit access, and technical support. Community-based organizations and innovation networks (ASORECH, CDRO, ADIPAZ, LTACs) will act as local anchors, ensuring legitimacy, adoption, and sustainability. In addition, the mentioned institutions implement specific programs for women and youth, ensuring that social inclusion remains a core perspective.

Financial viability at scale will be achieved through a blended model: (i) national public financing aligned with climate, water, and soil policies; (ii) international climate and development finance (e.g., IADB-led GCF facility in Central America, [Colombia's GCF project on CSA](#), RAMP-Accelerate); and (iii) cost-recovery and

reinvestment mechanisms through producer cooperatives and microfinance institutions, particularly for water infrastructure and soil conservation services.

This multi-actor, multi-finance strategy ensures that scaling is not donor-dependent but embedded in institutional systems and market mechanisms, creating pathways for long-term sustainability and resilience.

- **Partnerships & Leverage** – Identify internal CGIAR champions and external partners that have a stake in championing the innovation bundle or package, provide evidence of demand (surveys, requests for collaboration, alignment with policies or strategies) and commitment (MOUs, LOIs, budget allocations from partners, or other evidence of demand), and specify major non-CGIAR programs or investments this track can leverage (budgets, partners, policies, programs, etc). (~300 words)

The scaling of climate-smart water and soil solution bundles will be anchored in CGIAR's Climate Action and Digital Transformation programs. Climate Action provides participatory approaches such as PICSA and LTACs, which will be directly applied as a delivery mechanism that facilitates bundles adaptation with farmers and local institutions. In parallel, the Digital Transformation Accelerator supports the uptake of these solutions by embedding them into digital tools—such as agroclimatic platforms and mobile applications—that can extend access, improve usability, and enhance decision-making at scale. Together, these programs ensure that innovations are not developed in isolation but embedded within established CGIAR platforms, connected to national technical staff, and aligned with ongoing regional efforts, enabling scaling in Guatemala and Colombia while creating spillover potential to other Latin American countries and facilitating south-south exchange with other countries in Africa and Asia.

At the local level, there is evidence of both demand and capacity for scaling. **ASORECH** has demonstrated both local demand and technical capacity to adopt and scale participatory climate-smart water–soil management practices. Since 2019, over 4,000 families (43,000 people in total, including people with disabilities and their caregivers) have been reached, validating technologies and strengthening community organization. Partners such as **CDRO** in Totonicapán and Quetzaltenango and **ADIPAZ** in Zacapa, have replicated these bundles with around 400 families each, showing adaptability across diverse contexts. These commitments are backed by collaboration agreements, shared budget allocations, and formal requests from municipalities seeking to expand coverage.

Scaling is further supported through alignment with **national policies and programs**. In Guatemala, bundles contribute to the *Framework Law on Climate Change*, the *National Climate Change Policy*, the *National Climate Change Action Plan*, the *National Water Policy*, the *National Irrigation Policy*, and the *National Drought Strategy*. In Colombia, bundles align with the *Climate Change Law*, *National Integrated Water Resource Management Policy*, the *Sustainable Soil Management Policy*, the *National Adaptation Plan*, and the *Food and Nutrition*

Security Policy. Also, in both countries the bundles will contribute to the achievement of international commitments, such as the NDCs.

- **Risks & Bottlenecks** – Identify specific risks (financial, political, operational, social, or others) and bottlenecks (policy, institutions, finance, technology) in the specific context and place that could hinder scaling. Describe potential mitigation strategies to address these risks and bottlenecks. (~250 words)

The scaling of climate-smart water and soil solutions in Guatemala and Colombia may face a series of risks and bottlenecks that could hinder progress. **Political risks** include the frequent turnover of decision-makers and shifting government priorities, which often disrupt program continuity and delay the integration of innovations into public agendas. To reduce this risk, the strategy seeks to secure commitments at both national and local levels, and to anchor practices in existing policies and legal frameworks, providing stability against political cycles. Building strong relationships with the **technical staff of partner institutions** is equally critical, as their lower turnover ensures continuity in implementation and creates trusted champions for sustained adoption.

Financial risks stem from budget cuts and competition for limited resources, which may weaken the capacity of public institutions and local governments. To mitigate this, the approach emphasizes diversifying funding sources and leveraging solid evidence of impact to attract new investments from both public and private actors.

At the **operational level**, social unrest—such as strikes, blockages, or protests—can disrupt field activities, particularly in areas with fragile governance. Mitigation measures include working through local organizations with legitimacy on the ground, adopting flexible planning, and allocating contingency resources.

Finally, **structural bottlenecks** persist, including limited coordination among public institutions and the digital divide that restricts timely access to information in rural areas. These will be addressed by strengthening innovation networks that connect diverse actors and promoting locally adapted tools and methodologies. Together, these strategies reduce risks and bottlenecks, creating more stable conditions for responsible and sustainable scaling.

Section 3 – Responsible Scaling Trajectory & Achievable Scaling Target

Purpose: To define what can realistically be achieved with S4I support in the next 2–3 years — including achievable reach, milestones, resources, activities, and safeguards for responsible scaling.

- **Readiness & Responsible Scaling** – Provide evidence of scaling readiness (IPSR or similar evidence). Describe actions for ensuring that scaling will be inclusive (extending to women, youth, marginalized groups) and environmentally sustainable. (~150 words)

The innovation has a Scaling Readiness level of 9, demonstrating maturity and proven impact under field conditions. Evidence from Guatemala and Colombia

shows that climate-smart soil and water bundles generate tangible results, including higher yields, improved water availability during dry seasons, and reduced climate vulnerability. Farmers report benefits in income, food security, and resilience, supported by positive cost–benefit analyses of practices such as terraces, live barriers, and community reservoirs. Specific adoption rates and quantitative results can be found in the evidence section. The Innovation Use level is 8, since the innovations are already being used by end-users non directly involved in its development through the implementation of community led projects in other municipalities in both Guatemala and Colombia.

Scaling will be inclusive by integrating gender and youth-responsive design, leveraging participatory platforms (PICSA, LTACs, MyGeoFarm), and ensuring accessibility for marginalized groups. Partner networks further reinforce this approach, as they include institutions with dedicated programs for women and youth, embedding social inclusion as a central dimension of the innovation pathway. It will be **responsible** by advancing climate-smart soil–water ecosystem solutions only to levels that do not create risks of environmental degradation, combining farm- and watershed-level interventions that conserve water, restore soils, protect biodiversity, and strengthen resilience without overburdening natural systems.

- **Achievable target, Resources & Activities** – Define what can realistically be achieved in 2–3 years starting 2026 forward with S4I support and current/planned partnerships. Outline the key resources (staff, institutional, financial, technical) and activities (e.g., training, policy engagement, distribution) required. (~250 words)

With S4I support, existing partnerships, and an estimated USD 500,000 per year (2026–2028), the initiative will engage 20,000 producers and their families by 2026, ensuring at least 40% participation of women and youth. This will be achieved through inclusive training programs, participatory platforms (PICSA, LTACs, CSVs), and technical assistance delivered with local cooperatives, NGOs, and agribusiness partners. At the same time, the partner network will expand by at least 12 new institutions (ministries, lenders, producer organizations) to strengthen institutional capacity for scaling. The nested network approach will allow continuous social capital building that will enable an accelerated scaling in the subsequent years.

By 2027, coverage will extend to 50,000 farmers, supported by MEL systems capturing quantitative outcomes and farmers' perceptions of reduced crop losses, improved food security, and adaptive capacities. Resources will be mobilized as follows: CGIAR staff expertise for training and MEL design; institutional partners for extension, credit, and outreach; financial resources for scaling infrastructure such as water systems and soil conservation practices; and technical inputs from Climate Action SP and the Digital Transformation Accelerator to integrate digital tools. Activities will include dissemination of bundles through participatory approaches and extension systems, deployment of mobile tools, and structured policy engagement to embed solutions into national frameworks.

By 2028, the initiative aims to reach 80,000 farmers, with MEL documenting 15–20% perceived reductions in crop losses and at least one national or regional policy incorporating the bundles. In this way, scaling will be consolidated as an institutional and financial strategy, embedded within national systems and positioned for sustained impact.

- **Timeline, Milestones & Monitoring** – Provide an estimate of annual milestones for 2026–2028 (users reached, partnerships formed, policies adopted). Explain how progress will be monitored, lessons captured, and adaptations made. (~200 words)

The scaling pathways are structured around critical milestones that mark progress toward sustainable adoption in Guatemala and Colombia. By 2026, the initiative will reach 20,000 farming families, with at least 40% women and youth. Territorial networks of cooperatives, extension agents, and local organizations will be consolidated as anchors, while at least 12 new strategic partners (ministries, cooperatives, NGOs, lenders) will be engaged. Participatory design of initial scaling pathways will ensure local ownership and suitability.

By 2027, coverage will expand to 50,000 farmers, supported by validated participatory scaling pathways implemented through territorial networks. A first formal agreement with a ministry or donor program will embed bundles into extension or financing mechanisms. Monitoring, Evaluation, and Learning (MEL) will document improvements in food security, climate resilience, and adaptive capacities, while farmer-to-farmer exchanges and digital tools expand adoption.

By 2028, the initiative will reach 80,000 farmers, with scaling pathways consolidated as institutional roadmaps and at least one national or regional policy formally adopting the bundles. MEL will capture evidence of a 15–20% perceived reduction in crop losses and income stability. Progress will be monitored through quantitative adoption metrics and qualitative farmer perceptions, complemented by annual reflection workshops that ensure lessons are captured, strategies adapted, and accountability maintained.

References

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**Optional Annexes****1. Partner commitments (MOUs, LOIs, letters of support)**

No.	Partner	Type	Evidence
1	Asociación Regional Campesina Ch'ortí (ASORECH)	NGO	Link
2	Asociación de Desarrollo Integral de la Parte Alta de Zacapa (ADIPAZ)	NGO	Link
3	Asociación de Cooperación para el Desarrollo Rural de Occidente (CDRO)	NGO	Link
4	Agropecuaria Popoyán	Private sector	Link
5	Municipality of Olopa	Public sector	Link

2. Melia

No	Type	Topic	Year of publication	Link
1	Case study	Strengthening capacities for the scaling of Local Technical Agroclimatic Committees through collaborative action	2025	https://cgspace.cgiar.org/items/0145af70-eafe-4cc7-abe4-0c6327bf53bc
2	Case study	Scaling Agroclimatic Services: Reaching 810,000 farmers with Climate-Resilient Solutions across three Latin-American countries	2025	https://cgspace.cgiar.org/server/api/core/bitstreams/6da384c1-566f-46a3-a34f-a1a4c77542fb/content
3	Manual	Implementation Manual for Climate-Smart Agriculture (CSA) Practices	2025	https://cgspace.cgiar.org/bitstreams/a57cbe38-9289-4155-a671-32a96ada7df2/download
4	Report	Monitoring of Ecosystem-based Adaptation (EbA) and Climate-Smart Agriculture (CSA) practices in the TeSAC of Olopa	2024	https://cgspace.cgiar.org/server/api/core/bitstreams/b76e9de9-31bc-4fdc-b854-c7ea51207b8b/content
5	Article	Climate-smart agriculture reduces capital-based livelihoods vulnerability: evidence from Latin America	2024	https://www.frontiersin.org/journals/sustainable-food-systems/articles/10.3389/fsufs.2024.1363101/full
6	Brief	The Climate-Smart Villages (CSV) of Olopa as a Local Innovation Platform for Scaling Ecosystem-Based Adaptation in Guatemala.	2024	https://cgspace.cgiar.org/server/api/core/bitstreams/160c5b45-bf01-4731-99e2-37c21b9174c1/content

No	Type	Topic	Year of publication	Link
7	Report	Monitoring and evaluation of the implementation of Participatory Integrated Climate Services for Agriculture (PICSA) in communities in eastern Guatemala, within the framework of the PRO-Resiliencia program.	2024	https://cgspace.cgiar.org/items/402625d5-0376-4308-bce8-c5e8624d6f83
8	Case study	Co-innovation and scaling: Climate-smart agriculture country profiles	2023	https://cgspace.cgiar.org/server/api/core/bitstreams/cb598032-745f-41ee-b9a6-24bab52738e3/content
9	Report	Adoption of Climate-Smart Agriculture practices in the Climate-Smart Village of Guatemala: insights by gender and household characteristics	2019	https://cgspace.cgiar.org/server/api/core/bitstreams/2b2312f6-758f-4680-a22e-3909e611f9a7/content
10	Article	Costs and benefits of climate-smart agriculture: The case of the Dry Corridor in Guatemala	2017	https://www.sciencedirect.com/science/article/pii/S0308521X16301160